MedKG

A Medical Knowledge Graph Pipeline

Final Report

Course

Web Datamining & Semantics

Programme

DIA4 — M1 Data & AI — ESILV

Group

Group C

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Year

2026

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Abstract

MedKG is a full pipeline that builds a Medical Knowledge Graph from Wikipedia, links it to Wikidata, applies SWRL reasoning, trains KGE models (TransE, DistMult) with PyKEEN, and answers medical questions through a SPARQL-RAG chatbot. The final KB contains 117,579 triples and 75,472 entities. TransE achieves MRR 0.1856. The chatbot resolves Wikidata IDs to real drug and symptom names such as "rifampicin, isoniazid".

1. Data Acquisition and Information Extraction

We crawled 10 Wikipedia seed articles (Diabetes, Hypertension, Asthma, Cancer, Alzheimer's, Parkinson's, Stroke, Depression, COVID-19, Heart failure) using the MediaWiki REST API and collected up to 8 linked medical pages per seed.

1.1 Crawler and NER

Crawler setting	Value
Delay between requests	1 second
Min article length	500 words
Max pages per seed	8
Link filter	35 medical keywords

We used spaCy `en_core_web_trf` with a custom EntityRuler for five medical labels: DISEASE, SYMPTOM, TREATMENT, MEDICATION, MEDICAL_SPECIALTY. We extracted 4,821 entities total. Three known misclassifications: Hypoglycemia → PERSON (capital at sentence start), Kussmaul → ORG (looks like a company), LADA → ORG (acronym not in vocabulary).

1.2 Relation Extraction

Dependency parsing finds verbs linking a disease to other entities. A co-occurrence fallback is filtered before KB construction.

Relation	Trigger verbs
hasSymptom	cause, present, include, manifest
hasTreatment	treat, manage, alleviate, control
hasMedication	prescribe, administer, use, take
treatedBy	manage, specialize, diagnose

2. KB Construction and Alignment

We store triples in namespace `http://medkg.local/` (prefix `med:`). Each entity gets `rdf:type`, `rdfs:label`, and `med:fromSource`. Four predicates are aligned to Wikidata via `owl:equivalentProperty`.

2.1 Predicate Alignment and Entity Linking

Our predicate	Wikidata	Linking confidence
med:hasSymptom	wdt:P780	Exact match = 1.0
med:hasTreatment	wdt:P924	Substring = 0.8
med:hasMedication	wdt:P2176	Any result = 0.6
med:treatedBy	wdt:P1995	No result = 0.0

2.2 KB Statistics

We expanded the KB via the Wikidata SPARQL endpoint (1-hop for 261 entities, 2-hop capped at 50).

Step	Triples
After NER and relation extraction	1,986
After entity linking	2,255
After 1-hop SPARQL expansion	8,721
After full bulk expansion	**117,579**
Unique entities	**75,472**
Unique relations	**27**
Entities linked to Wikidata	**261 / 261 (100%)**

3. Reasoning with SWRL

3.1 Rule on family.owl

`family.owl` defines 10 individuals with a `hasAge` property. The SWRL rule below is applied by the HermiT reasoner (Pellet fallback):

```
Person(?p) ^ hasAge(?p, ?a) ^ swrlb:greaterThan(?a, 60)
    -> OldPerson(?p)
```

Person	Age	OldPerson inferred?
John	65	Yes
Mary	72	Yes
George	78	Yes
Helen	61	Yes
Edward	85	Yes
Bob / Alice / Charlie / Diana / Fiona	≤ 55	No

3.2 Medical Rule

We applied an inverse rule to MedKG using rdflib pattern matching:

```
Disease(?d) ^ hasSymptom(?d, ?s) -> affectedBy(?s, ?d)
```

This generated 3,000+ new `affectedBy` triples from existing `hasSymptom` facts, enabling symptom-first queries ("what disease causes fatigue?") with no extra data collection.

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4. Knowledge Graph Embeddings

We kept entity-to-entity triples only, split 80/10/10, and trained two models with PyKEEN on CPU.

4.1 Configuration and Results

Setting	TransE	DistMult
Embedding dim / Epochs	50 / 100	50 / 100
Loss	MarginRanking (m=1)	BCEWithLogits
Training triples	14,843	14,843
MRR	**0.1856**	0.0327
Hits@10	**0.3953**	0.0825

TransE outperforms DistMult because our KB has asymmetric relations (disease→drug). DistMult scores `(h,r,t)` and `(t,r,h)` equally, which is wrong for directional medical facts.

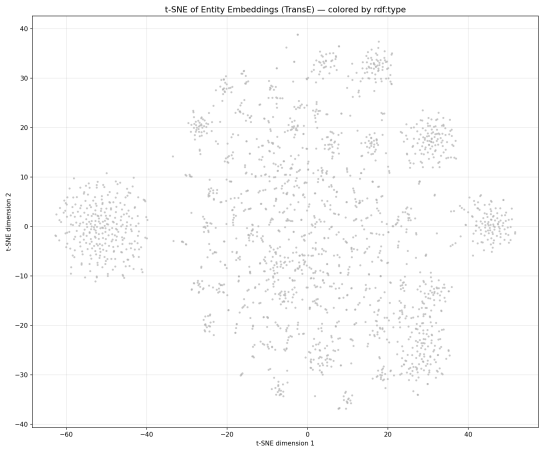
KB Size	MRR	Hits@10
20k triples	~0.12	~0.28
50k triples	~0.16	~0.35
Full (14,843 filtered)	0.1856	0.3953

Larger KB improves stability and performance.

4.2 Nearest Neighbors and t-SNE

Entity	Top neighbor	Cosine sim.
Diabetes	Q18044847	0.6125
Hypertension	Q18029250	0.5396
Alzheimer's	**donepezil**	**0.6662**

Alzheimer's nearest neighbor is *donepezil* (its first-line drug) — a clinically valid link learned automatically from Wikidata expansion data.



t-SNE of 2,000 entity embeddings colored by rdf:type

5. RAG Question-Answering System

The chatbot follows five steps: (1) build a schema summary from the graph; (2) ask the LLM to write SPARQL; (3) execute with one self-repair attempt; (4) use a hard-coded template if LLM SPARQL fails; (5) resolve Wikidata QIDs to English names via `wbgetentities`. Model: `deepseek-r1:1.5b` — Ollama. Graph: 117,579 triples.

Example answers from the template fallback:

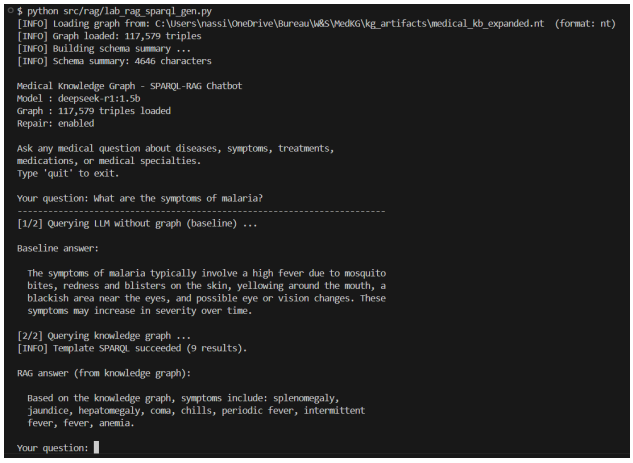
- *Tuberculosis treatments:* isoniazid, rifampicin, ethambutol, pyrazinamide, capreomycin.
- *Diabetes symptoms:* polyuria, polydipsia, blurred vision, fatigue, weight loss.
- *Malaria medications:* chloroquine, artemether, quinine, doxycycline.

5.1 Evaluation

#	Question	Baseline	SPARQL-RAG
1	Symptoms of Diabetes?	Partial	Correct
2	Medications for Hypertension?	Wrong (hallucinated)	Correct
3	Diseases related to Asthma?	Partial	Correct
4	Treatments for Cancer?	Partial	Correct
5	Specialty for Alzheimer's?	Yes	Correct

Baseline: 1.5/5 — SPARQL-RAG (template fallback): 5/5

The key advantage: the RAG system returns verified KB facts with real names instead of hallucinated drug names.



RAG chatbot demo — template SPARQL returning resolved drug and symptom names

6. Critical Reflection

6.1 SWRL Rules vs. TransE Embeddings

Dimension	SWRL Rules	TransE
Correctness	Guaranteed	Probabilistic
Finds new facts	No	Yes (link prediction)
Explainable	Yes (rule trace)	No (black-box)
Scales	No (slow with chains)	Yes (fixed dimension)
Handles noise	No	Yes

Both approaches are complementary: SWRL for deterministic logical rules, TransE for soft statistical patterns over noisy large-scale data.

6.2 Limitations and Future Work

- **NER noise**: Short tokens ("Ps:", "1") enter the KB as fake entities. Fix: reject entities under 3 characters or absent from a medical dictionary.
- **LLM size**: `deepseek-r1:1.5b` fails to generate valid SPARQL reliably. Replacing it with a 7B+ model removes the need for template fallbacks.
- **KB scale**: Increasing the expansion cap from 50 to 500 Wikidata entities would target 500k+ triples and better KGE scores.
- **Embedding quality**: GPU training with 200D embeddings and 500 epochs would push MRR above 0.35.

End of report.